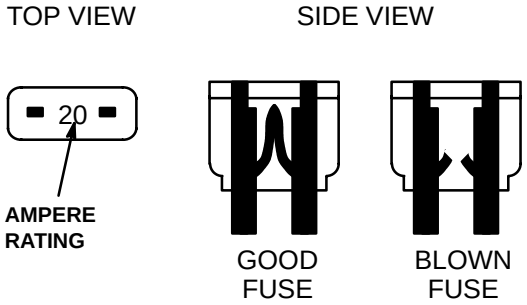
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fuse links, maxi-fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

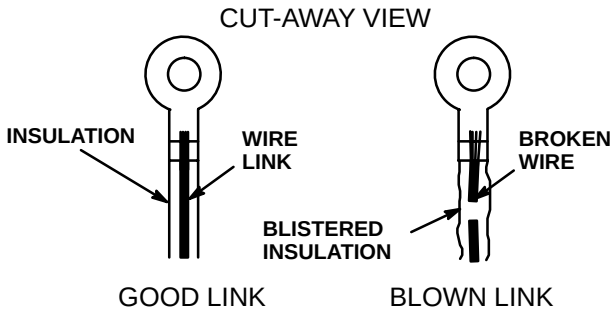


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSE LINK



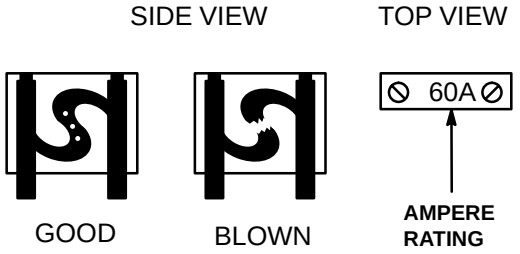
Fuse links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fuse link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fuse link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; If the fuse link stretches, the wire has melted.

When replacing fuse links, first cut the protected wire where it is connected to the fuse link. Then, tightly crimp or solder the new link to the protected wire.

Fuse links are often identified by color coding of the insulation, as shown here:

FUSE LINK COLOR CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

MAXI-FUSE



Maxi-fuses have a transparent colored plastic housing. To check a maxi-fuse, look at the fuse element through the side of the housing.

To replace a maxi-fuse, pull it from the fuse box or panel. Always replace a blown maxi-fuse with a new one having the same ampere rating.

The ampere rating of a maxi-fuse can also be determined by following the color code shown here:

MAXI-FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
20	Yellow
30	Light Green
40	Amber
50	Red
60	Blue

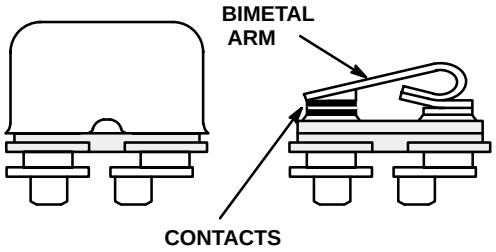
CIRCUIT BREAKER

Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

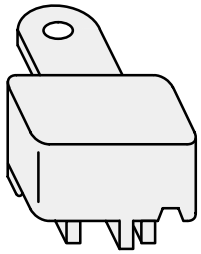
Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



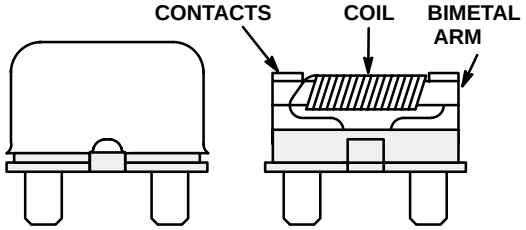
IN-LINE MOUNTED CYCLING TYPE



In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

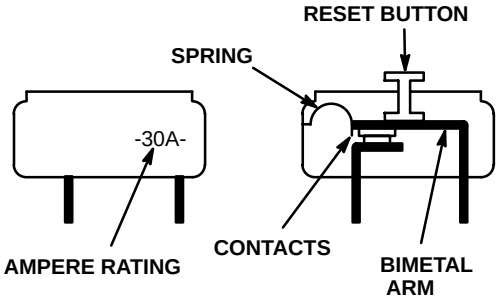
FUSE PANEL MOUNTED NON-CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



FUSE PANEL MOUNTED MANUAL RESET TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)

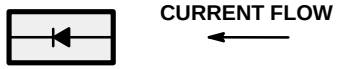


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE



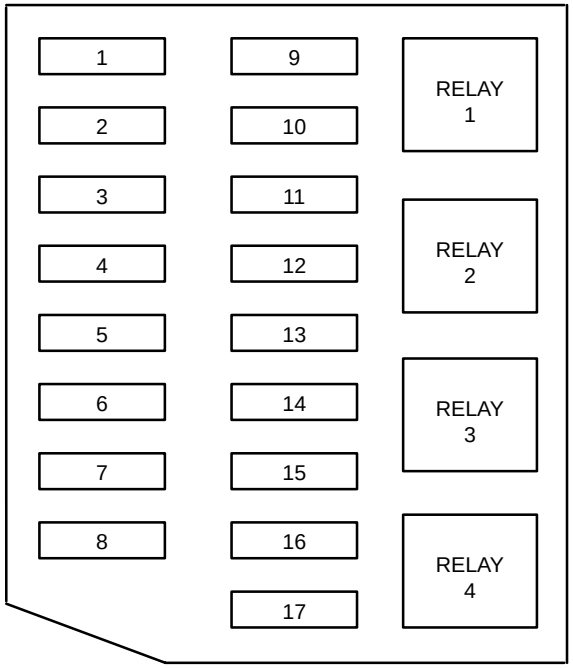
Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

AMPERE RATING	HOUSING COLOR
BLADE TYPE FUSE	
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green
MAXI-FUSE	
20	Yellow
30	Light Green
40	Amber
50	Red
60	Blue

11-3 FUSE PANEL/CIRCUIT PROTECTION

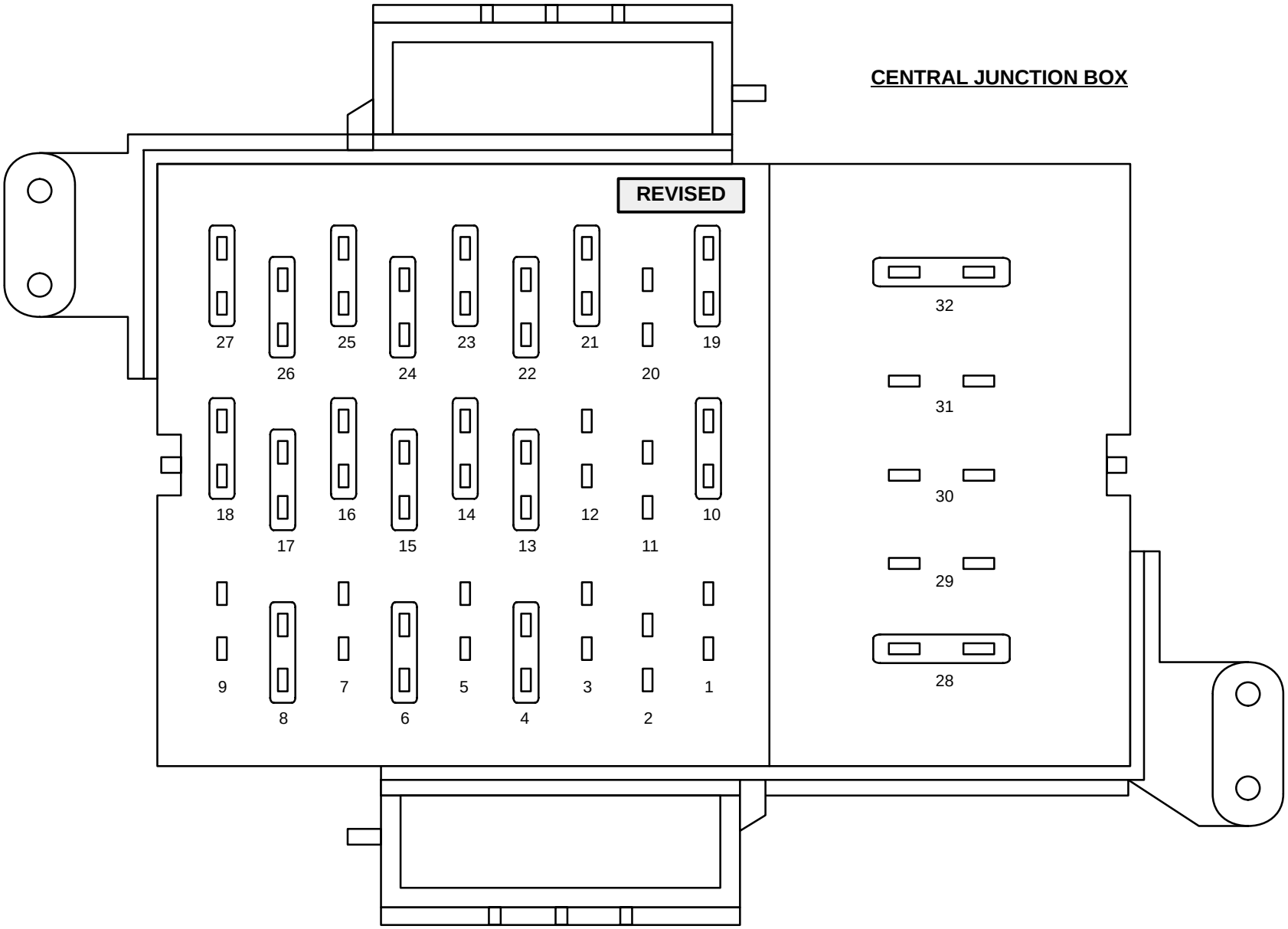
* WITH POLICE

BATTERY JUNCTION BOX



RELAY NUMBER	NAME
1	Rear Defrost Relay
2	Horn Relay
3	Cooling Fan Relay
4	Air Suspension Low Speed Relay
*4	Police Power Relay

MAXI-FUSE	AMPS	CIRCUITS PROTECTED	PAGE
8	30	Air Suspension System	13-4
9	50	Ignition Switch Feed to Central Junction Box Fuses 16, 18, 32	13-7
10	50	Ignition Switch Feed to Central Junction Box Fuses 4, 6, 8, 13, 15, 17, 22, 28	13-8
11	50	Central Junction Box Fuses 1, 21, 23, 25, 27	13-6
12	30	PCM, Data Link Connector, PCM Power Relay, Natural Gas Vehicle Module	13-5
13	50	Cooling Fan High Relay	13-1
14	40	Rear Window Defrost, Central Junction Box Fuse 10	13-6
15	40	Anti-Lock Brake System (ABS) Module	13-5
16	50	Police Option Fuse Holder	13-2
FUSE	**AMPS**	**CIRCUITS PROTECTED**	**PAGE**
1	20	Fuel Pump Prime Connector, Fuel Pump Relay, Fuel Valve Relay	13-5
2	30	Starter Motor Relay, Generator, Ignition Switch Feed to Central Junction Box Fuse 14	13-7
3	25	Radio, Subwoofer Amplifier, CD Changer	13-2
4	30	Police Power Relay	13-2
5	15	Horn Relay	13-2
6	20	Daytime Running Lamps (DRL) Module	13-5
CIRCUIT BREAKER	**AMPS**	**CIRCUITS PROTECTED**	**PAGE**
7	20	Power Door Locks, Power Seats, Luggage Compartment Lid Release, Adjustable Pedal, Power Lumbar Seats and Power Reclining Seats	13-3
17	30	Cooling Fan Low Relay	13-6


CENTRAL JUNCTION BOX
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11-5 FUSE PANEL/CIRCUIT PROTECTION

CENTRAL JUNCTION BOX

Maxi-Fuse	Amps	Circuits Protected	Page
28	20	Power Windows	13–8
29	–	Not Used	11–4
30	–	Not Used	11–4
31	–	Not Used	11–4
32	20	ABS Valves	13–7, 13–15
Mini-Fuse	**Amps**	**Circuits Protected**	**Page**
1	–	Not Used	13–6
2	–	Not Used	13–8
3	–	Not Used	13–6
4	10	Air Bags	13–8
5	–	Not Used	13–6
6	15	Instrument Cluster, Warning Lamps Module, Transmission Control Switch (TCS), LCM	13–8
7	–	Not Used	13–6
8	25	PCM Power Relay, Coil-On-Plugs, Radio Noise Capacitor	13–8
9	–	Not Used	13–6
10	10	Rear Window Defrost	13–6
11	–	Not Used	13–8
12	–	Not Used	13–7
13	5	Radio	13–8
14	10	Traction Control Switch, ABS, Instrument Cluster	13–7
15	15	Speed Control Servo, Main Light Switch Illumination, LCM, Clock, Police Power Relay	13–8
16	15	Reversing Lamps, Turn Signals, Shift Lock, DRL Module, EVO Steering, Electronic Day/Night Mirror	13–7
17	30	Wiper Motor, Wiper Control Module	13–8
18	30	Heater Blower Motor	13–7
19	20	PowerPoint	13–6

CENTRAL JUNCTION BOX, cont.

Mini-Fuse	Amps	Circuits Protected	Page
20	–	Not Used	13–8
21	15	Multifunction Switch, Lighting Control Module (LCM), PATS Indicator	13–6
22	15	Speed Control DEAC Switch, Hazard Lights, Brake Pedal Position (BPP) Switch	13–8
23	15	Power Windows/Door Locks, PATS, Exterior Rear View Mirrors, EATC Module, Instrument Cluster, Clock	13–6
24	10	Left Low Beam Headlamp	87–2
25	20	Power Point, Cigar Lighter, Emergency Flashers	13–6
26	10	Right Low Beam Headlamp	87–2
27	20	LCM, Main Light Switch	13–6